



mcp-jev

Minimal [Model Context Protocol](#) server that exposes TypeSafe **Jev** (System One) structured decisions to Cursor and other MCP clients.

Jev is the fast classifier; your main LLM is the writer. This server is the bridge — one tool call in, typed probabilities out.

Requirements

- Docker and Docker Compose
- A TypeSafe API key from console.typesafe.ai

For development and tests only: Python 3.11+

Quick start (Docker)

Copy `.env.example` to `.env` and set your key (local only — never commit):

```
copy .env.example .env
```

Start the MCP HTTP API:

```
docker compose up --build -d
```

The server listens at `http://127.0.0.1:8000/mcp` (Streamable HTTP transport).

MCP client setup

Example configs for Cursor, Claude Code, and Claude Desktop are in `samples/`. Copy the one you use — client configs are not committed (only examples).

Cursor (quick start):

```
mkdir .cursor
copy samples\cursor.mcp.json.example .cursor\mcp.json
```

Start the container before using MCP (`docker compose up -d`). Reload MCP servers after changing `.env`.

See `samples/README.md` for Claude Code and Claude Desktop paths and notes.

Local development (optional)

```
python -m venv .venv
.venv\Scripts\activate # Windows
pip install -e ".[dev]"
pytest
```

Run without Docker (stdio transport, for MCP Inspector):

```
set JEV_TRANSPORT=stdio
python -m mcp_jev.server
```

Or HTTP on the host (matches Docker behavior):

```
set JEV_TRANSPORT=streamable-http
set JEV_HOST=127.0.0.1
python -m mcp_jev.server
```

Tool: `jev_decide`

Evaluate a `state` against one or more typed `questions` using TypeSafe Jev. Returns probabilities and structured answers — not natural language.

Parameter	Type	Required	Description
<code>state</code>	string, object, or array	yes	Content Jev judges. Prefer a JSON object with named fields.

Parameter	Type	Required	Description
questions	object	yes	Map of question id → question definition.
model	string	no	Default <code>jev-latest</code> . Pin e.g. <code>jev-1.13.0</code> when tuning thresholds.

Question types

Type	Purpose	Required fields
noul	Yes/no probability	instructions; optional criteria { "true", "false" }
choice	Pick one option	instructions, criteria (map option → description)
score	Ordered scale	instructions, criteria (array, low → high, 2–10 levels)

Example

```
{
  "state": {
    "message": "My card was charged twice for order A-104."
  },
  "questions": {
    "is_billing": {
      "type": "noul",
      "instructions": "Is `message` about billing or payments?"
    },
    "department": {
      "type": "choice",
      "instructions": "Which team should handle `message`?",
      "criteria": {
        "billing": "Charges, invoices, refunds",
        "technical": "Bugs or outages",
        "other": "None of the above"
      }
    }
  }
}
```

Success response (pretty-printed JSON string):

```
{
  "model": "jev-1.13.0",
  "answers": { },
  "usage": { "input_tokens": 0, "output_tokens": 0 }
}
```

On failure, the tool returns a structured error object instead of throwing:

```
{
  "error": true,
  "status": 422,
  "message": "Validation failed: ...",
  "details": { }
}
```

Environment variables

Variable	Required	Default	Description
TYPESAFE_API_KEY	yes	—	Bearer token from TypeSafe console
JEV_TRANSPORT	no	stdio (streamable-http in Docker)	MCP transport
JEV_HOST	no	127.0.0.1 (0.0.0.0 in Docker)	HTTP bind address
JEV_PORT	no	8000	HTTP port
JEV_HTTP_PATH	no	/mcp	Streamable HTTP endpoint path
JEV_STATELESS_HTTP	no	true	Stateless HTTP sessions
JEV_DEFAULT_MODEL	no	jev-latest	Default model for tool calls
JEV_API_BASE	no	https://api.typesafe.ai	API base URL (testing/mocking)
JEV_TIMEOUT_SEC	no	30	Per-request HTTP timeout
JEV_MAX_RETRIES	no	3	Retries for HTTP 429/529

Limits

Documented by TypeSafe; lightly validated locally in v1:

Limit	Value
State + all questions (combined)	~64,000 tokens
State + longest single question	~32,000 tokens
Choice options	up to 255
Score levels	2–10

Testing

```
pytest
```

Tests mock HTTP — no API key required.

Manual smoke test

1. Set `TYPESAFE_API_KEY` in `.env`.
2. Run `docker compose up --build -d`.
3. Open Cursor and call `jev_decide` with one `noul` question on a short string.
4. Confirm `answers.*.noul` is a float in `[0, 1]`.

Security

- Never log or commit `TYPESAFE_API_KEY`.
- Treat `state` as potentially sensitive — nothing is persisted in v1.
- Threshold and routing logic belong in the agent prompt, not in this server.

References

- [TypeSafe — Introducing System One and Jev](#)
- [MCP Python SDK](#)
- Project spec: [docs/spec.md](#)